

Chapter 1 — Introduction, Literature Review and Data

Continuous-Time Modelling of Wind Speed for Derivatives Pricing and Risk Management

1 Motivation and Background

Wind is not a tradable asset. It cannot be stored, its delivery cannot be guaranteed, and no counterparty can be compelled to supply it on demand. Yet an entire class of energy-market participants, from onshore wind producers and transmission system operators to portfolio managers hedging renewable-heavy generation mixes, carries direct financial exposure to its fluctuations. This is the defining tension of weather-linked risk management: the underlying quantity that determines cash flow is physical and non-tradable, while the instruments used to hedge it must nonetheless be priced, marked to market, and settled in monetary terms. Wind speed is, in this sense, a demanding test case for derivative pricing methodology, because it resists the standard toolkit built for tradable underlyings.

For a German onshore wind producer, this exposure has two distinct legs. The first is volumetric: low-wind periods reduce generation directly, and because wind speed is positively autocorrelated at short lags and seasonally heteroskedastic rather than i.i.d., a run of calm days is not diversified away by the following week. The second is a price risk correlated with, rather than independent of, the first: German day-ahead electricity prices exhibit a merit-order effect, in which episodes of high aggregate wind generation push low-marginal-cost supply onto the grid and depress the clearing price when producers are generating the most. A storm system moving across the North Sea can lift aggregate German wind output to a large share of instantaneous demand within hours, driving the day-ahead price toward zero or, on occasion, below it, while onshore producers are running near full capacity.

Because revenue is the product of quantity and price, this negative correlation is partially self-offsetting at the revenue level: it dampens revenue variance relative to the independent case rather than compounding it. That natural hedge is, however, both weak and asymmetric. It is weak because the empirical correlation between German onshore wind generation and the day-ahead price is only $\rho = -0.16$ over 2015–2024 (Chapter 3), reflecting the many other drivers of the clearing price. It is asymmetric because it erodes the producer’s upside, capture prices falling when output is highest, without placing a floor under the downside: a sustained low-wind episode is not reliably compensated

by a price move of matching magnitude. What remains after the natural hedge is an uncompensated volumetric exposure, and it is that residual leg which a derivative written on wind speed itself, rather than on the electricity price alone, is designed to address.

The economic stakes are not abstract. To illustrate the order of magnitude involved, consider a representative 100 MW onshore wind farm operating at the capacity factor computed for the Nordfriesland site in Section 4.2 (25.70%, from the Weibull-based annual energy production baseline; this is a gross resource figure, and a real project would forgo a further 8–15% to wake losses and availability). Annual generation is then $100 \text{ MW} \times 8,760 \text{ h} \times 0.257 \approx 225,100 \text{ MWh}$. The mean German day-ahead price computed directly from the SMARD dataset used in this thesis (Section 4.3) differs sharply across three windows: €34.60/MWh for the 2015–2020 subsample, €126.47/MWh for the 2021–2024 subsample (inflated by the European energy-price crisis), and €71.38/MWh over the full 2015–2024 sample. Applying these three prices to the illustrative farm gives an annual revenue of roughly €7.8 million, €28.5 million, and €16.1 million respectively. These figures are offered strictly as an order-of-magnitude illustration, built transparently from a stated capacity factor and observed price data, not as a precise valuation of any actual asset. Against a revenue stream of this size, a partial mispricing of a hedging instrument written against it is economically material long before one reaches portfolio scale: as reported in Section 2, calibrating the same pricing model on a misaligned proxy dataset misprices the variance-swap fair strike by 88.0% even after correcting for measurement height.

Two features of the German market make it a natural setting for this study. First, Nordfriesland, on Germany’s North Sea coast in Schleswig-Holstein, is the country’s highest-density onshore wind corridor, and its geographic centroid is a reasonable proxy for the aggregate wind generation fleet reported by the German transmission system operators’ joint platform, SMARD (Bundesnetzagentur, smard.de). That alignment is not merely asserted: regressing daily SMARD onshore generation on the cube of Nordfriesland hub-height wind speed yields $R^2 = 0.59$, against $R^2 = 0.0001$ for the identical regression run on the Bologna proxy (Chapter 3). Second, the European Energy Exchange (EEX) lists the most liquid German power options market in Europe, providing at least the theoretical possibility of extracting a market-implied price of risk for wind-linked instruments. That possibility is more remote than data access alone would suggest: EEX options are written on the power price, not on wind speed or wind generation, so recovering a wind risk premium from them would require a joint model of price and generation capable of separating the wind premium from the power-price premium. As discussed in Section 3.6 and revisited under the $\theta = 0$ benchmark assumption in Chapter 2, no such option-implied calibration is attempted in this thesis. Together, these two facts motivate treating Nordfriesland hub-height wind as the reference underlying for a derivatives-pricing exercise grounded in German electricity market structure, rather than as an arbitrary meteorological time series.

The modelling programme that this thesis reports on was not designed around Nordfriesland data from the outset. It began, for reasons of data availability, on a

deliberately low-cost proxy: four years of 10 m wind speed observations for Bologna, Italy, obtained via the OpenWeatherMap API (Section 4.1). Every methodological step, from the Box–Cox transformation and the AR(4)/CAR(4) mean dynamics to the GARCH(1,1) volatility layer and the stochastic-volatility embedding, was built, tested, and validated on this proxy dataset before being applied, unchanged in its functional form, to the Nordfriesland reference data. This two-dataset structure is deliberate rather than incidental: it allows the thesis to separate two questions that are usually conflated in applied derivatives work, namely whether the *methodology* is correct and whether the *data* it is fed are fit for purpose. Chapter 3 turns this separation into a quantitative argument, the Value of Information, that measures the pricing error attributable to using a geographically and physically misaligned proxy instead of an aligned reference dataset.

2 Research Questions and Contribution

The thesis is organised around three research questions.

- RQ1.** To what extent does a continuous-time autoregressive model of order four, CAR(4), combined with a GARCH(1,1) conditional-volatility layer and calibrated on wind speed aligned in both geography and measurement height with the generation aggregate being hedged, yield a tractable closed-form fair strike for a synthetic wind variance swap?
- RQ2.** What is the Value of Information, defined here as the relative mispricing in the model-implied variance-swap fair strike attributable to proxy misalignment, from upgrading a geographically and physically misaligned proxy dataset (Bologna, 10 m) to a reference dataset aligned with the German onshore wind fleet (Nordfriesland, 100 m hub height), measured both before and after a wind-profile height correction?
- RQ3.** How does a Cumulative Accumulated Wind Speed (CAWS) instrument behave as a left-tail complement to the variance swap, in the sign and dispersion of its payoff, when evaluated on serially uncorrelated annual observations?

RQ1 is addressed methodologically in Chapter 2, which develops the full modelling pipeline from raw wind speed to a continuous-time stochastic-volatility representation, and applied in Chapter 3, which uses the resulting parameters to price a synthetic variance swap under a piecewise-constant-volatility convention built on the GARCH-in-wind-speed specification of Tol [1997]. RQ2 and RQ3 are addressed in Chapter 3, where the cross-dataset validation exercise (Diebold–Mariano forecast tests, the Value of Information comparison, and the CAWS hedging simulation) is carried out on both datasets side by side.

The thesis makes three contributions, corresponding one-to-one with the three gaps identified in Section 3.7. First, it calibrates the CAR(4)+GARCH pricing framework on wind speed that is simultaneously at turbine hub height and geographically aligned

with the traded generation aggregate being hedged, a combination absent from the wind-derivatives literature reviewed below. Second, it produces an explicit, quantified Value of Information estimate for that alignment decision, expressed both as a raw pricing gap and, after applying a wind-profile height correction, as the gap that survives height adjustment. That residual gap is itself the informative quantity: it shows that geographic climate mismatch, not measurement height alone, is the dominant source of proxy-data pricing error. Third, it constructs and stress-tests a CAWS-based instrument addressing the left tail of wind-linked revenue risk, the sustained low-wind episodes to which a variance swap is by construction indifferent, evaluated on non-overlapping annual observations to avoid the serial-correlation bias that overlapping rolling windows would otherwise introduce into the risk metrics. That evaluation choice trades statistical precision for freedom from serial-correlation bias: it leaves four annual observations for Dataset A and ten for Dataset B, so the resulting tail statistics are reported as descriptive rather than inferential, and the cross-dataset comparison rests on the sign and dispersion of the payoff rather than on a 5% quantile estimated from four points.

The principal findings can be stated at the outset. The model-implied fair strike for the synthetic wind variance swap is $K_{\text{var}}^B = 0.873$ on the Nordfriesland reference dataset against 0.054 on the Bologna proxy, a raw relative mispricing of 93.8% that falls only to 88.0% once the $1.931\times$ wind-profile height correction is applied. The $16.1\times$ ratio between the two strikes factorises cleanly into $1.82\times$ attributable to the GARCH conditional-variance state $h(t_0)$ and $8.85\times$ attributable to the seasonal variance level C_0 , which locates the mispricing in the climatology of the site rather than in its volatility state, and is the quantitative basis for the claim that geography dominates height. On forecast accuracy, all four fitted specifications reject the persistence benchmark decisively (Diebold–Mariano statistics of 8.26 on Dataset A and 16.76 on Dataset B) while being observationally equivalent to one another in one-step point-forecast terms, a result developed in Section 5 and Chapter 3.

3 Literature Review

The methodology developed in this thesis draws on six distinct strands of literature, corresponding to the five successive layers of the modelling pipeline developed in Chapter 2 and the pricing framework applied in Chapter 3: the marginal statistical properties of wind speed, discrete-time forecasting models, conditional-volatility modelling, continuous-time and Lévy-driven processes, stochastic-volatility option-pricing theory, and, finally, weather and energy derivatives pricing under an explicit market price of risk. This section reviews each strand in turn and closes with a statement of the specific gap this thesis addresses.

3.1 Statistical Properties of Wind Speed

The starting point for any statistical treatment of wind speed is the observation that it is a positive, right-skewed, seasonally heteroskedastic quantity, and therefore not amenable

to Gaussian time-series methods without a preliminary transformation. Tuller and Brett [1984] provide the classical physical justification for the marginal distribution most commonly used to describe it. Their argument proceeds from first principles: if the two horizontal components of the wind velocity vector are independently and identically normally distributed with zero mean and equal variance (a reasonable approximation for wind in the free atmosphere, away from surface friction effects), then the magnitude of the vector, which is what an anemometer records, follows a Weibull distribution with shape parameter $k = 2$, the Rayleigh special case. More generally, departures from this idealised isotropy produce Weibull-distributed speeds with k in the neighbourhood of two but not necessarily equal to it. This is a physical derivation, not merely a curve-fitting convention, and it explains why the Weibull distribution has become the default parametric choice in wind-resource assessment.

Garcia et al. [1998] test this claim empirically across twenty meteorological stations in Navarre, Spain, comparing Weibull and lognormal fits to hourly-mean wind speed via a log-log linearisation of the Weibull cumulative distribution function (the “Weibull probability paper” method) and nonlinear regression for the lognormal alternative. They find the Weibull distribution provides a materially better fit at essentially all stations, with R^2 values in the high-0.90s versus the high-0.80s for the lognormal, and they use the fitted distributions to estimate wind-farm energy production, establishing the now-standard link between the marginal distribution and the integral of a turbine power curve against its density. Their result is a useful benchmark against which to read the distributional analysis conducted on the two datasets used in this thesis (Section 4): the Weibull distribution is confirmed as the better fit for the hub-height Nordfriesland reference dataset, consistent with Garcia et al. [1998] and Tuller and Brett [1984], whereas the 10 m Bologna proxy dataset is better described by a lognormal distribution, a departure from the Weibull consensus that Section 4.1 attributes to the proxy’s low measurement height, where surface-layer friction and diurnal boundary-layer effects violate the isotropy assumption underlying the Rayleigh–Weibull derivation.

Because neither marginal family is Gaussian, a variance-stabilising transformation is required before the Gaussian time-series machinery of the following sections can be applied. The Box–Cox power transformation serves this role throughout the thesis, with the transformation parameter λ estimated by profile likelihood on each dataset and then rounded to a value admitting the closed-form pricing formulas of Section 3.6; the specific values adopted, and the cost of the rounding, are reported in Sections 4.1 and 4.2.

A separate physical literature governs the comparison of wind speeds measured at different heights, which the Value of Information exercise of Chapter 3 requires. Peterson and Hennessey [1978] assess the power-law profile $u(z) \propto z^\alpha$ against measured profiles and conclude that an exponent of $\alpha = 1/7$ is adequate for realistic but conservative estimates of available wind power, except at extremely rough sites. This is the exponent adopted for the single height correction applied in this thesis, giving a variance scaling of $(100/10)^{2/7} \approx 1.931$ between 10 m and 100 m. It is important to be clear that α is an empirical parameter rather than a physical constant: it varies with surface roughness

and atmospheric stability, with values nearer 1/10 typical over open water and larger values over rough terrain, so a coastal German site sits near the middle of the plausible range. The sensitivity of the correction to this choice is treated explicitly in Chapter 3, where the correction is confined to the cross-dataset comparison and applied nowhere else.

3.2 Time-Series Models and Long Memory

Once the marginal distribution and the deterministic seasonal component have been dealt with, the residual dynamics of wind speed remain strongly autocorrelated, and a substantial literature has developed autoregressive and long-memory models to capture this. Torres et al. [2005] fit month-specific ARMA models to hourly wind speed at five Navarre stations after a Weibull-shape-based power transformation and per-hour standardisation, and show that ARMA forecasts significantly outperform the naive persistence benchmark, particularly beyond the first few forecasting horizons, although the improvement narrows as the horizon lengthens. This paper establishes both the transform-then-model pipeline adopted throughout the present thesis and a natural point of comparison for the incremental value of the AR(4) model over persistence reported in Chapter 2.

A parallel strand of the literature asks whether wind speed exhibits long memory (autocorrelation that decays hyperbolically rather than exponentially), which would argue for a fractionally integrated ARFIMA specification rather than a finite-order AR or ARMA model. Haslett and Raftery [1989], in one of the earliest and most influential papers in this tradition, model daily wind speed across twelve Irish synoptic stations by combining spatial kriging with a fractionally differenced temporal component on the square-root-transformed series, and show that naive short-memory estimators materially understate the standard errors of long-run wind-power assessments once long-range dependence is properly accounted for. Kavasseri and Seetharaman [2009] apply fractional ARIMA models directly to day-ahead wind speed forecasting at four North Dakota sites, reporting an average 42% reduction in forecast mean squared error relative to a persistence benchmark and consistent gains over standard (short-memory) ARIMA. Ailliot et al. [2006] take a different route to capturing structure beyond a fixed-coefficient AR model, proposing a Markov-switching autoregression with time-varying spatial coefficients to track the translation of synoptic weather systems across a wind field, calibrated on ECMWF North Atlantic hindcast data. Their application is spatial rather than the single-location setting used in this thesis, but the underlying idea, that AR coefficients should be allowed to vary with the state of the system rather than held fixed, foreshadows the seasonally modulated and stochastic-volatility extensions developed in Chapter 2.

Taken together, this cluster motivates a specific modelling choice made in Chapter 2: an AR(4)/ARFIMA order-selection exercise is carried out on both datasets used in this thesis, following the diagnostic logic of Haslett and Raftery [1989] and Kavasseri and Seetharaman [2009], but the finite-order AR(4) specification is ultimately retained as the

operating model because it embeds directly into the continuous-time CAR(4) framework of Section 3.4 without requiring a separate fractional-integration apparatus, and because the long-memory evidence on the datasets in this thesis, reported in Chapter 2, does not decisively favour the more heavily parameterised alternative.

3.3 Conditional Volatility

The residuals of an AR model fit to deseasonalised wind speed are typically not homoskedastic: calm periods and volatile periods cluster in time, in the same way that volatility clusters in financial return series. The tools for modelling this clustering originate in econometrics. Engle [1982] introduces the autoregressive conditional heteroskedasticity (ARCH) process, in which the conditional variance of a serially uncorrelated, zero-mean process is itself an autoregressive function of past squared innovations; Engle motivates the model with UK inflation data and develops both a maximum-likelihood estimator and a Lagrange-multiplier test for the presence of ARCH effects that remains the standard pre-estimation diagnostic to this day. Bollerslev [1986] generalises ARCH to GARCH by allowing the conditional variance to depend on its own lagged values as well as on lagged squared innovations (the volatility analogue of extending an AR model to an ARMA model), and shows that a low-order GARCH(1, 1) specification typically achieves the same fit as a much higher-order ARCH model with far fewer parameters. GARCH(1, 1) is the specification adopted for the conditional-volatility layer $h(t)$ throughout this thesis.

Tol [1997] is the paper that transplants this machinery specifically into wind speed modelling. Working with daily wind data from Shearwater, Canada, Tol fits Gamma-AR(2)-GARCH(1,1) and Normal-AR(2)-GARCH(1,1) specifications and demonstrates that the heteroskedastic model outperforms a homoskedastic AR-only benchmark, with the ARCH effect noticeably stronger in the underlying wind velocity components than in the scalar wind speed series itself. This is the direct precedent for the two-component volatility decomposition used throughout this thesis, $\sigma_{\text{total}}^2(t) = \sigma_{\text{seasonal}}^2(t) \times h(t)$, in which a deterministic Fourier-based seasonal variance is multiplied by a GARCH(1, 1) conditional-variance factor. It is also the source of the two-component volatility specification on which the pricing convention adopted in Chapter 3 rests. Tol's contribution is econometric rather than one of derivatives pricing: the paper contains no variance swap and no fair-strike formula, and it is cited here for establishing $h(t)$ as the conditional-variance state variable for wind speed. Following that treatment, the present thesis freezes h at its pricing-date value $h(t_0)$ over the life of the contract, a piecewise-constant simplification in the spirit of the piecewise-monthly seasonal variance of Alaton et al. [2002], and derives the resulting fair strike $K_{\text{var}}^B = h \times C_0$ directly from the Fourier decomposition of $\sigma_{\text{seasonal}}^2(t)$, whose first and second harmonics integrate to zero over a full year.

3.4 Continuous-Time and Lévy-Driven Processes

Discrete-time AR and GARCH models are estimated on daily data, but a derivative contract must in general be priced at an arbitrary maturity that does not fall on a daily grid point, which motivates embedding the discrete-time dynamics into a continuous-time stochastic process. Brockwell and Marquardt [2005] provide the mathematical foundation for this step, developing the theory of Lévy-driven continuous-time ARMA (CARMA) processes: state-space representations built around a companion matrix A and driven by an arbitrary second-order Lévy process rather than Brownian motion, which allows the resulting marginal distributions to be asymmetric and heavy-tailed rather than Gaussian. They further construct a fractionally integrated extension (FI-CARMA) by convolving the CARMA kernel with a Riemann–Liouville fractional-integration kernel, connecting the long-memory literature reviewed in Section 3.2 to the continuous-time state-space formalism used in Chapter 2’s CAR(4) construction: the companion matrix A , its eigenvalues, and the matrix exponential $\exp(A\tau)$ that propagates the conditional mean forward in time are all direct applications of this framework.

Schwartz [1997] supplies the connection to commodity-derivatives pricing specifically. Schwartz compares one-, two-, and three-factor models of commodity spot prices, respectively a simple mean-reverting (Ornstein–Uhlenbeck) process, a model adding a stochastic convenience yield, and a model further adding stochastic interest rates, and evaluates their implications for the term structure of futures prices and for hedging performance. The single-factor Ornstein–Uhlenbeck specification in Schwartz’s framework is, in essence, the $p = 1$ special case of the CAR(p) family used in this thesis, and its treatment of mean reversion under a risk-neutral measure is a direct precedent for the wind-futures pricing framework developed by Benth and Šaltytė Benth [2009] and adopted, under the $\theta = 0$ benchmark described in Section 3.6, in Chapter 2.

Barndorff-Nielsen and Shephard [2001] extend the non-Gaussian Ornstein–Uhlenbeck idea to stochastic volatility directly, constructing a volatility process driven by a positive Lévy subordinator (the background driving Lévy process) rather than a Brownian motion, so that returns become Gaussian mixtures capable of reproducing heavy tails and volatility clustering while retaining closed-form results for integrated volatility and its temporal aggregation. This paper is the theoretical bridge between the GARCH-based discrete-time volatility proxy $h(t)$ used in Chapters 2–3 of this thesis and the continuous-time variance process $V(t)$ introduced in the stochastic-volatility extension of Chapter 2: it demonstrates formally that a GARCH-type discrete filter can be understood as a discretisation of an underlying continuous semimartingale, which is precisely the conceptual step motivating the two-SDE system developed there.

3.5 Stochastic Volatility

Heston [1993] is the canonical reference for stochastic-volatility option pricing and supplies the mathematical template for the two-SDE system used in the theoretical stochastic-volatility chapter of this thesis. Heston models the variance of an asset’s

return as a square-root (Cox–Ingersoll–Ross-type) process correlated with the return innovation itself, and derives a closed-form European option price via Fourier inversion of the joint characteristic function rather than by solving the pricing partial differential equation directly. This technique generalises immediately to other payoffs, including the variance-swap-style contract priced in Chapter 3 of this thesis. Heston’s framework requires two conditions to be satisfied for the model to be well posed: the Feller condition, which guarantees that the variance process remains strictly positive almost surely, and a specification of the market price of volatility risk connecting the physical and risk-neutral dynamics of the variance process. Chapter 2 of this thesis adopts Heston’s two-SDE structure, a wind-speed level process and a CIR-type variance process coupled through the Feller condition, as a theoretical (not estimated) extension of the GARCH-based volatility model, for the reasons discussed under the gap statement in Section 3.7.

3.6 Weather and Energy Derivatives, and the Market Price of Risk

The applied weather-derivatives literature translates the statistical machinery reviewed above into pricing formulas for non-tradable underlyings, and it is here that the question of the market price of risk becomes unavoidable. Alaton et al. [2002] provide an accessible template: they fit an Ornstein–Uhlenbeck process with piecewise (monthly) seasonal variance to Stockholm temperature data, estimate it via martingale estimating functions applicable to discretely observed diffusions, and price temperature-index derivatives (heating- and cooling-degree-day contracts) both by an analytic approximation and by Monte Carlo simulation. Because temperature, like wind speed, is not a traded asset, the pricing measure cannot be pinned down by a replication argument in the usual Black–Scholes sense, and Alaton, Djehiche and Stillberger introduce an explicit, generally non-zero market price of risk parameter to complete the model in this incomplete-market setting.

Benth and Šaltytė Benth [2009] extend this OU-based approach specifically to wind speed, embedding a Box–Cox-transformed, seasonally adjusted daily wind speed series into a $CAR(p)$ state-space model with seasonal mean and seasonal volatility, calibrated on New York wind data using an $AR(4)$ specification with Box–Cox parameter $\lambda \in \{0.1, 0.2\}$. They derive closed-form wind futures prices as conditional expectations of the CAR state vector under a chosen pricing measure, and, notably, they show that the classical Samuelson effect, the tendency of futures price volatility to increase as the contract approaches maturity, can break down or reverse for autoregressive orders $p > 1$, a consequence of the long memory embedded in a higher-order autoregressive structure that has no analogue in the single-factor Schwartz-type model. This paper is the direct methodological template for the $CAR(4)$ construction, the Box–Cox transformation, and the Euler-discretisation link between the discrete $AR(4)$ and continuous-time $CAR(4)$ representations used throughout Chapter 2 of this thesis.

The question of how to specify the market price of risk θ in this incomplete-market setting is addressed empirically by two further papers. Härdle and López Cabrera [2009] infer the market price of weather risk implicitly from Chicago Mercantile Exchange-

traded Berlin cumulative-temperature futures, modelling both piecewise-constant and time-varying specifications, and find that the implied market price of risk is significantly different from zero and displays a clear seasonal pattern. Dorfleitner and Wimmer [2010] conduct a related empirical exercise on CME temperature futures directly, comparing detrended and non-detrended index models against observed market prices and showing that a simple non-detrending specification tracks quoted futures prices closely, that weather forecasts materially move prices up to eleven days ahead of settlement, and that a mismodelled seasonal trend produces systematic, exploitable mispricing across the winter and summer contract cycle. Both papers demonstrate that, where reliable option-implied or futures-implied price data exist, the market price of risk for a weather-linked underlying is empirically estimable and generally non-zero.

Alexandridis and Zapranis [2013] return to wind speed specifically and are the direct source for the two index conventions used in the pricing exercise of this thesis. They model daily wind speed with a non-parametric wavelet-network layer superimposed on a mean-reverting, seasonally adjusted Ornstein–Uhlenbeck process, benchmark it against the linear CAR-type approach of Benth and Šaltytė Benth [2009], and find that the wavelet-network specification materially improves in- and out-of-sample fit. More directly relevant to this thesis, they derive closed-form futures pricing formulas for two wind indices: the Nordix index and the Cumulative Accumulated Wind Speed (CAWS) index (referred to as “cumulative average wind speed” in that paper), the latter of which is adopted, in the normalised rolling-annual-mean form defined in Section 4 and Chapter 3, as the left-tail hedging instrument evaluated in this thesis. In practical terms, the instrument pays out whenever the realised cumulative average wind speed over the contract period falls short of a pre-agreed strike level, so that a sustained run of calm weather, precisely the condition that depresses a producer’s physical generation, triggers a compensating cash inflow that offsets the associated shortfall in electricity revenue.

Finally, Carr and Lee [2009] provide the theoretical foundation for variance-swap replication that motivates, without directly supplying, the implemented pricing formula used in Chapter 3 of this thesis. Their review formalises the classical result that a variance swap can be synthetically replicated by a static position in a log-contract combined with dynamic delta-hedging in the underlying, quantifies the higher-order replication error incurred when this hedge is only discretely rebalanced, and shows that the floating leg of a continuously monitored variance swap corresponds to a $1/K^2$ -weighted portfolio of out-of-the-money options across the strike surface: the same logic that underlies the construction of the VIX index.

This replication argument requires a liquid market in options on the underlying, which does not exist for German wind speed or wind generation. The only actively traded wind derivatives, the Nordix futures referenced by Alexandridis and Zapranis [2013], are written on Norwegian rather than German wind, and no options market exists for the Nordfriesland-aligned SMARD generation series used in this thesis. In the absence of this option-implied information, the variance swap priced in Chapter 3

does not attempt Carr–Lee-style replication; instead, it follows the closed-form, model-implied formula of Tol [1997], $K_{\text{var}}^B = h \times C_0$, under the explicit benchmark assumption $\theta = 0$ described in Chapter 2.

This benchmark is adopted deliberately, as a tractable simplification given the absence of EEX option-implied data for this specific underlying, and not as a claim that the market price of risk for wind-linked instruments is in fact zero. The evidence from Härdle and López Cabrera [2009] and Dorfleitner and Wimmer [2010], reviewed above for a related but distinct underlying, would in any case not support such a claim. Standard risk-aversion arguments suggest a directional hypothesis worth stating explicitly: if wind-linked risk carries a premium comparable to that documented for temperature, producers seeking protection against low-wind revenue shortfalls would be expected to accept, and counterparties to demand, compensation above the physical-measure expectation $h \times C_0$, so that a true market-implied strike would plausibly lie *above* rather than below the $\theta = 0$ benchmark used here; this is a hypothesis about the direction of the omitted risk premium, not an estimate of its magnitude. Carr and Lee’s framework remains, in this thesis, the designated theoretical motivation for a future extension in which θ is calibrated to EEX German power options, rather than the model actually estimated.

3.7 Gap Statement

Read together, this literature establishes a well-developed toolkit: Weibull/lognormal marginal fitting, AR/ARFIMA mean dynamics, GARCH conditional volatility, $\text{CAR}(p)$ continuous-time embedding, Heston-type stochastic volatility, and OU-based weather-derivatives pricing under an explicit market price of risk. Three gaps in that toolkit remain, and this thesis addresses each of them directly.

First, the wind-derivatives literature reviewed in Section 3.6 is calibrated either on temperature (a more heavily studied underlying with a longer exchange-traded history) or on wind speed measured at a location and height that need not correspond to the generation asset actually being hedged. Benth and Šaltytė Benth [2009] and Alexandridis and Zapranis [2013] both calibrate on New York data; neither is calibrated on wind speed that is simultaneously (i) measured at or near turbine hub height and (ii) geographically aligned with a specific, actively traded generation aggregate. This thesis addresses that gap directly by using ERA5 reanalysis wind speed at 100 m hub height over the Nordfriesland corridor, the geographic centroid of the SMARD German onshore wind generation series against which the pricing exercise in Chapter 3 is ultimately validated.

Second, no paper in this review quantifies, in the same modelling framework, the pricing cost of using a geographically or physically misaligned proxy dataset instead of an aligned one. The comparative exercises in that literature, whether across volatility specifications (Section 3.3) or across long-memory orders (Section 3.2), are concerned with comparing model specifications on a fixed dataset, not with comparing the same model specification across two datasets of differing quality. Chapter 3’s Value of Information exercise, pricing the identical $\text{CAR}(4)+\text{GARCH}$ variance swap on both the

Bologna proxy and the Nordfriesland reference dataset and measuring the resulting gap before and after a wind-profile height correction, is designed specifically to fill this gap, and constitutes the thesis’s primary methodological contribution.

The physical reasoning behind that gap deserves stating, because it is what the height-corrected residual is designed to isolate. Nordfriesland’s open North Sea exposure brings frequent Atlantic storm passages and, at 100 m, a flow largely detached from surface obstacles. Bologna’s 10 m measurement sits inside the surface friction layer of the Po Valley, a basin enclosed by the Alps and the Apennines whose sheltered climatology suppresses both mean wind speed and the frequency of high-wind episodes. These are differences in baseline turbulence and storm climatology, not in measurement height, and a power-law profile correction rescales the level of the proxy series without reproducing them, which is why a substantial gap survives the correction in Chapter 3.

Third, a variance swap is by construction indifferent to the direction of the underlying’s movement: it pays on realised variance, and a sustained calm period, the single event a wind producer most needs protection against, registers as low variance rather than as a loss. The left tail of wind-linked revenue risk is therefore not addressed by the instrument that the literature reviewed in Section 3.6 prices, and although Alexandridis and Zapranis [2013] derive futures prices for the CAWS index, no study evaluates a cumulative-wind instrument as a left-tail complement to a variance swap on observations free of the serial correlation that its own rolling-window construction induces. Chapter 3 constructs and stress-tests exactly that pairing, on non-overlapping annual windows.

Two constraints bound what follows, and are stated here rather than left for the reader to infer. The stochastic-volatility literature (Section 3.5) generally estimates the variance-process parameters $(\kappa_V, \bar{v}, \xi, \rho)$ from either high-frequency data or option-implied information, neither of which is available here: the two datasets used in this thesis are daily, and no liquid options market exists on German wind speed or on the SMARD generation aggregate. Chapter 2 therefore develops the Heston-type stochastic-volatility system as a theoretical embedding of the GARCH proxy, verifying the Feller condition numerically and deriving the variance term structure in closed form, while explicitly deferring maximum-likelihood estimation of the full stochastic-volatility parameter set to a future extension using higher-frequency (e.g. hourly ECMWF) data. The same absence of option-implied data motivates the $\theta = 0$ benchmark of Section 3.6. Both constraints are limitations the thesis works within rather than gaps it closes, and both are revisited in the concluding synthesis of Chapter 3.

4 Data

This thesis draws on three data sources, used in complementary roles: two independent wind speed datasets (a proxy dataset on which the modelling methodology is built and validated, and a reference dataset on which the headline empirical results rest) and a market dataset of German electricity generation and prices used to construct the

synthetic derivative priced in Chapter 3. All three are described in full here, so that the later chapters can refer back rather than re-explaining them.

4.1 Dataset A: Bologna Proxy (10 m, 2015–2018)

Dataset A consists of hourly wind speed observations for the Bologna area, Italy (approximately 44.29°N, 11.20°E), obtained via the OpenWeatherMap API and distributed through the [beniamino98/greenfin-project](#) GitHub repository, at a nominal measurement height of 10 m. The raw file (`data/data.csv`) contains 35,064 hourly observations with no missing values, running from 23:00 UTC on 31 December 2014 to 22:00 UTC on 31 December 2018. Aggregation by arithmetic mean within each UTC calendar day yields $n = 1,462$ daily average wind speed observations: 1,460 complete twenty-four-hour days spanning 1 January 2015 to 30 December 2018, together with two partially populated boundary bins, one hour on 31 December 2014 and twenty-three hours on 31 December 2018, which arise because the raw sampling window is offset by one hour from the calendar-year boundary. The effective coverage is thus the four calendar years 2015–2018, and the two boundary bins are retained rather than trimmed so that the observation count matches the frozen Phase 1 output used throughout the thesis.

Table 1 reports the descriptive statistics computed for this series. Every diagnostic points to the same conclusion: raw daily wind speed at this site is strongly non-Gaussian. The mean exceeds the median by roughly 0.23 m/s, confirming a right-skewed distribution consistent with a physical lower bound near zero and an unbounded right tail driven by occasional storm days; skewness of +1.76 and excess kurtosis of +4.74 are both far from the Gaussian values of zero, and the Jarque–Bera statistic of 2,122 against a $\chi^2(2)$ critical value near 6 rejects normality overwhelmingly.

Table 1: Descriptive statistics, raw daily average wind speed, Dataset A (Bologna, 10 m, 2015–2018).

Statistic	Value	Note
n	1,462	daily observations
Mean	2.4687 m/s	
Median	2.2417 m/s	mean > median $\Rightarrow$ right skew
Std. deviation	0.9481 m/s	
Minimum / Maximum	0.9333 / 9.1125 m/s	no calm daily averages
Skewness	+1.7616	Gaussian = 0
Excess kurtosis	+4.7351	Gaussian = 0
Jarque–Bera p -value	< 0.000001	normality strongly rejected

A Weibull distribution fit by maximum likelihood gives shape $\hat{k} = 2.64$ and scale $\hat{c} = 2.77$ m/s, placing the shape parameter in the range of two to three typically reported in wind-resource assessment, above the Rayleigh value of $k = 2$ that the Tuller and Brett [1984] isotropy argument reviewed in Section 3.1 would imply; however, on an AIC/BIC comparison against a lognormal fit ($\hat{m} = 0.84$, $\hat{s} = 0.34$), the lognormal wins

decisively (log-likelihood $-1,730.6$ versus $-1,960.5$ for the Weibull, a gap of roughly 460 AIC units). This is a substantive departure from the Tuller and Brett [1984]–Garcia et al. [1998] consensus, and is attributable to the 10 m measurement height: the isotropy assumption behind the Rayleigh–Weibull derivation is a free-atmosphere approximation, progressively violated by surface friction and diurnal boundary-layer effects closer to the ground, and the lognormal’s thinner left tail better matches the empirical absence of near-zero daily averages at this height.

The Box–Cox transformation (Section 3.1) is applied to bring the series toward normality. The likelihood-maximising value is $\lambda_{\text{opt}} = -0.450$, which achieves near-perfect Gaussianisation (skewness -0.005 , excess kurtosis $+0.128$, Jarque–Bera $p = 0.606$); however, the closed-form pricing formulas used in later phases require $1/\lambda$ to be a positive integer, and -0.450 does not satisfy this constraint. Following Benth and Šaltytė Benth [2009], who adopt $\lambda \in \{0.1, 0.2\}$ for the same CAR(p) continuous-time embedding, $\lambda = 0.1$ (so that $1/\lambda = 10$) is adopted as the frozen working parameter for this dataset throughout the thesis, at the cost of leaving a residual right skew (skewness 0.601 , Jarque–Bera $p \approx 0$) that the seasonal-mean and seasonal-variance decomposition and the AR(4) fit of Chapter 2 subsequently reduce further. After the truncated Fourier decomposition into a seasonal mean $\Lambda(t)$ (annual harmonic dominant, $R^2 = 0.126$) and a seasonal variance $\sigma^2(t)$ (mean level $C_0 = 0.13138$, winter-to-summer variance ratio of approximately 6.4:1, $R^2 = 0.226$), the resulting standardised series $\tilde{W}(t)$ has mean 0.001 and standard deviation 0.967 , close to the zero-mean, unit-variance target, but retains significant serial autocorrelation and conditional heteroskedasticity (Ljung–Box $p < 0.001$ on both $\tilde{W}(t)$ and $\tilde{W}(t)^2$ at all tested lags), which motivates the AR(4) and GARCH(1,1) layers developed in Chapter 2.

Because Dataset A is recorded at 10 m rather than at a representative turbine hub height, and because its four-year window is short relative to the multi-decade samples used in the wind-derivatives literature reviewed in Section 3.6, it is treated throughout this thesis strictly as a *proxy*: a low-cost dataset on which every stage of the modelling pipeline is developed and validated before being applied, without re-specification, to the reference dataset described next.

4.2 Dataset B: Nordfriesland Reference (100 m, 2015–2024)

Dataset B consists of daily wind speed at 100 m hub height for Nordfriesland, Schleswig-Holstein, Germany (54.77°N , 8.85°E), obtained from the Open-Meteo Historical Weather API, which serves ERA5 reanalysis output. The 100 m variable (`wind_speed_100m`) is extracted directly from the reanalysis and requires no vertical-profile correction, since it is a direct model output rather than a surface-station reading extrapolated to hub height. Two companion series are retained for validation and future regime analysis: `wind_speed_10m`, used as a consistency check against the primary series, and `wind_direction_100m`, retained for possible directional-regime analysis beyond the scope of this thesis. The sample spans ten full calendar years, 2015–2024, comprising 3,653 daily observations, which is the exact day count of that window given the three

leap years it contains.

Table 2 reports the corresponding descriptive statistics. The mean wind speed of 8.07 m/s is roughly 3.3 times the Dataset A mean, consistent with the combined effect of hub height and a substantially windier geographic location; skewness (+0.585) and excess kurtosis (+0.111) are both markedly closer to Gaussian values than Dataset A’s, though the Jarque–Bera test still rejects normality at conventional levels.

Table 2: Descriptive statistics, raw daily wind speed, Dataset B (Nordfriesland, 100 m, 2015–2024).

Statistic	Value	Note
n	3,653	daily observations
Mean	8.068 m/s	
Median	7.760 m/s	mean > median $\Rightarrow$ right skew
Std. deviation	3.236 m/s	
Minimum / Maximum	1.383 / 20.258 m/s	no calm daily averages
Skewness	+0.585	Gaussian = 0; cf. +1.762 Dataset A
Excess kurtosis	+0.111	Gaussian = 0; cf. +4.735 Dataset A
Jarque–Bera p -value	< 0.001	normality rejected

Consistent with the Tuller and Brett [1984]–Garcia et al. [1998] consensus, and in contrast with Dataset A, the Weibull distribution ($\hat{k} = 2.67$, $\hat{c} = 9.09$) fits Dataset B better than the lognormal on both AIC and BIC. The Box–Cox likelihood is maximised at $\lambda_{\text{opt}} = 0.404$; the tractable value adopted for closed-form pricing is $\lambda = 0.5$ (so $1/\lambda = 2$), a much smaller rounding distance than the one forced on Dataset A, reflecting the higher hub-height mean wind speed’s less extreme skew. The Fourier seasonal-mean fit is again dominated by the annual harmonic ($R^2 = 0.103$), and the seasonal-variance fit yields a mean level $C_0 \approx 1.163$ with $\sigma^2(t)$ ranging over $[0.944, 1.463]$, a winter-to-summer variance ratio of approximately 1.55:1. Expressed on the same basis, the seasonal modulation at Nordfriesland is thus far calmer than the 6.4:1 ratio found for Bologna, though both datasets place peak variance in the winter months. The resulting standardised series $\tilde{W}(t)$ again passes its basic centring and scaling checks (mean ≈ 0 , standard deviation ≈ 1.005) while retaining significant autocorrelation in levels and in squares, motivating the same AR(4) and GARCH(1,1) layers applied to Dataset A. As an illustrative application, integrating the fitted Weibull density against the power curve of a representative 3 MW turbine at 100 m hub height yields an annual energy production of 6.753 GWh and a capacity factor of 25.70%, a figure consistent with onshore wind performance in the North Sea coastal region and the figure used for the illustrative revenue calculation in Section 1.

Dataset B is used throughout this thesis as the *reference* dataset: it is geographically aligned with the SMARD German onshore wind generation series described next, measured directly at hub height, and long enough (ten years) to support the out-of-time validation exercise reported in Chapter 3, in which full-sample parameters are assessed against a 2020–2024 hold-out without re-calibration on the restricted window. All headline empirical claims in this thesis (the fitted variance-swap fair strike, the Value of

Information estimate, and the CAWS hedging simulation) rest primarily on Dataset B, with Dataset A retained as the benchmark against which the Value of Information is measured.

4.3 SMARD Generation and EEX Day-Ahead Prices

The synthetic derivative priced in Chapter 3 is written on German onshore wind generation, not on wind speed directly, and requires two further data series: hourly wind onshore generation and hourly day-ahead electricity prices, both published by the German transmission system operators' joint platform SMARD (Bundesnetzagentur, smard.de) for the full 2015–2024 window. Generation is delivered across two files, `Actual_generation.1.csv` and `Actual_generation.2.csv`, and prices across `Day-ahead_prices.1.csv` and `Day-ahead_prices.2.csv`. All four files are semicolon-separated, UTF-8-with-BOM encoded, and use a comma as the thousands separator, with timestamps in the form "Jan 1, 2015 12:00 AM"; the relevant generation column is labelled `Wind onshore [MWh] Calculated resolutions`.

The day-ahead price series requires a merge across two market conventions, reflecting a structural break in the underlying market: prior to 1 October 2018, Germany, Austria, and Luxembourg formed a single day-ahead bidding zone, reported in the column `DE/AT/LU [€/MWh]`; from that date, Austria was separated into its own bidding zone following a market-coupling reform, and the German price is reported instead in the column `Germany/Luxembourg [€/MWh]`. The merge rule applied throughout this thesis takes the `Germany/Luxembourg` column from 1 October 2018 onward and the `DE/AT/LU` column before that date. Both series are aggregated from hourly to daily frequency: generation as the sum of the 24 hourly values (a flow quantity, MWh/day), price as the arithmetic mean of the 24 hourly values (a level quantity, €/MWh). Two filters are applied rather than imputing: days exhibiting more than 20% missing hourly observations are dropped, as are days recording zero aggregate generation, which are treated as reporting errors rather than as genuine calm. Neither filter removed any day from the 2015–2024 generation series, which is complete.

The resulting daily series (`smard_daily_phase6.csv`) covers 1 January 2015 to 31 December 2024, 3,653 daily generation observations, with a mean national wind onshore generation of 255,560 MWh/day (computed directly from the processed file). The day-ahead price series is four days shorter, beginning on 5 January 2015, as the SMARD price files carry no quotation for the first four days of the sample; price-dependent calculations therefore rest on 3,649 daily observations. The corresponding day-ahead price series has a markedly different character before and after the onset of the European energy-price crisis: the mean price over 2015–2020 is €34.60/MWh, versus €126.47/MWh over 2021–2024 and €71.38/MWh over the full ten-year sample. This is the same three-way split used for the illustrative revenue calculation in Section 1. This regime shift also delimits the practical scope of a purely volumetric hedge: because the wind-speed-based instruments priced in Chapter 3 respond only to the physical quantity of wind, not to the price level, they cannot on their own offset the price-driven component

of revenue risk that dominated the 2021–2024 window, and a producer holding only this hedge would have remained fully exposed to the energy-price crisis itself. This structural shift is a feature of the German electricity market over the sample period rather than a data artefact, and Chapter 3 checks explicitly that the merit-order relationship between generation and price (used to motivate wind derivatives as a producer hedge) is not an artefact of the pre/post-2018 bidding-zone split either, by comparing the correlation between generation and price separately on either side of the October 2018 structural break.

4.4 Cross-Dataset Comparison

Table 3 summarises the two wind speed datasets side by side. It is presented once here and referred back to whenever a later result is reported for both datasets.

Table 3: Cross-dataset comparison: Dataset A (proxy) versus Dataset B (reference).

	Dataset A	Dataset B
Source	OpenWeatherMap API (beniamino98/ greenfin-project)	Open-Meteo Historical Weather API, ERA5 reanalysis
Location	Bologna, Italy (44.29°N, 11.2°E)	Nordfriesland, Germany (54.77°N, 8.85°E)
Height	10 m	100 m
Period	2015–2018 (4 yr, 1,462 obs)	2015–2024 (10 yr, 3,653 obs)
Primary variables	wind_speed (m/s)	wind_speed_100m, wind_speed_10m, wind_direction_100m
Advantages	Freely available; easy to download; sufficient for methodology development and validation	Hub-height (no profile correction needed); geographically aligned with SMARD generation; 10-year window enables out-of-sample validation; ERA5 reanalysis quality
Limitations	Proxy-quality, with less transparent provenance than a reanalysis product; 10 m height, for which the power-law profile correction recovers only a small part of the gap to hub height; low-wind site; short (4-year) window	Model-based reanalysis rather than direct anemometer observation; no sub-daily resolution used in this thesis

One derived series used in Chapter 3 is constructed from the two wind datasets documented above and is defined here alongside them. The Cumulative Accumulated Wind Speed index is taken in normalised rolling-annual-mean form,

$$\text{CAWS}_t = \frac{1}{252} \sum_{s=t-252}^t W(s), \quad (1)$$

where $W(s)$ is the daily mean wind speed in m/s: `wind_speed` for Dataset A and `wind_speed_100m` for Dataset B. Normalising by the 252-day window rather than

accumulating raw sums keeps the index in m/s and therefore comparable across the two measurement heights, at mean levels of roughly 2.47 m/s for Bologna and 8.07 m/s for Nordfriesland. The strike K_{CAWS} is the full-sample mean of $CAWS_t$, and the instrument pays $N \times (K_{CAWS} - CAWS_t)$, so that positive payoffs coincide with sustained low-wind conditions. As noted in Section 2, the risk metrics for this instrument are computed on non-overlapping annual observations, four for Dataset A and ten for Dataset B, even though the rolling series of equation (1) is what the time-series plots display.

The role of the cross-dataset comparison is not merely descriptive. It is the empirical premise for the Value of Information argument developed in Chapter 3: because the two datasets are processed through an *identical* modelling pipeline, any difference in the resulting derivative price can be attributed to the data themselves (their geographic location, height, and sample length) rather than to a difference in model specification. Table 3 is therefore the reference point against which that argument is built, motivating the upgrade from Dataset A to Dataset B that this thesis both executes and quantifies.

5 Thesis Outline

The remainder of the thesis is organised into two further chapters, followed by a concluding synthesis that closes Chapter 3 rather than standing as a separate chapter.

Chapter 2, *Methodology and Empirical Wind-Speed Modelling*, develops the modelling pipeline whose literature foundations are reviewed in Section 3, phase by phase, presenting Dataset A and Dataset B side by side at each stage: deseasonalisation and the Box–Cox transformation; the AR(4) conditional-mean model and its long-memory diagnostics; the GARCH(1,1) conditional-volatility layer; the continuous-time CAR(4) embedding, including its companion matrix, stationarity properties, and Normal-Inverse-Gaussian tail extension; and, finally, the theoretical stochastic-volatility extension that embeds the GARCH proxy into a Heston-type two-SDE system under the $\theta = 0$ benchmark. Dataset B is treated as the primary result throughout, with Dataset A retained as the validation benchmark on which each modelling choice is first tested.

Chapter 3, *Derivatives Pricing, Risk Management and Conclusions*, applies the calibrated model to answer the three research questions posed in Section 2.

It first prices the synthetic wind variance swap on both datasets, distinguishing the market-implied Track A fair strike, the 252 $\times$ -annualised realised variance of SMARD generation log-returns, from the model-implied Track B fair strike $K_{var}^B = h \times C_0$. The two tracks are computed under different annualisation and unit conventions: Track A is annualised and expressed in the units of generation log-returns, while Track B is left un-annualised in standardised wind units. Their raw numerical gap, roughly two orders of magnitude, is an artefact of those conventions rather than a measure of model error, and the two are reconciled by a delta-method rescaling before any comparison is drawn.

It then evaluates the forecasting models of Chapter 2 against the persistence benchmark using Diebold–Mariano tests [Diebold and Mariano, 1995, Harvey et al., 1997]. The result is a binary split rather than a graded hierarchy: every fitted specification,

from AR(4) through CAR(4) with Normal-Inverse-Gaussian innovations, improves on persistence at any conventional significance level, while being observationally equivalent to the others in one-step point-forecast terms. That equivalence is itself informative. It establishes that the models are distinguished not by conditional-mean accuracy but by their conditional-variance and tail representations, which is precisely why the thesis evaluates them on pricing and hedging criteria rather than on root mean squared error alone.

Chapter 3 then computes the Value of Information from the Dataset A-to-B upgrade, both before and after a wind-profile height correction restricted strictly to that comparison, and evaluates the CAWS instrument of Alexandridis and Zapranis [2013] as a left-tail complement to the variance swap on the non-overlapping annual basis defined in Section 4.4. It closes with a synthesis section that states the limitations of the $\theta = 0$ benchmark and of the daily-frequency stochastic-volatility identification problem, and outlines the option-implied and higher-frequency extensions left to future work.

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